

Dittert's conjecture in dimensions 11, 12, and 13

An exact computer-assisted boundary-exclusion proof

Working note

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Status. This is a reproducible working proof, not a peer-reviewed publication. All dimension-specific comparisons are checked with exact rational/integer arithmetic by two independent programs. The proof depends on published structural and permanent-minimization theorems cited below.

Abstract

For a nonnegative $n \times n$ matrix A whose entries sum to n , let r_i, c_j be its row and column sums and set

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \text{per } A.$$

Dittert's conjecture asserts that $\Phi(A) \leq 2 - n!/n^n$, with equality only at the uniform matrix J_n/n . We prove this for $n = 11, 12, 13$. The main device is the largest scalar λ for which A/λ is doubly superstochastic. A cut attaining λ is either marginal, in which case λ is controlled by the smallest row or column sum, or proper, in which case tightness forces a rectangular zero block in a dominated doubly stochastic matrix. The first case uses an exact lower bound on the two-independent-zero face; the second uses Hwang's staircase-face theorem. The remaining inequalities reduce to finitely many rational comparisons and three univariate Sturm calculations.

1 Statement and published inputs

Let

$$K_n = \left\{ A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j} a_{ij} = n \right\}, \quad \gamma_n = \frac{n!}{n^n}.$$

For $A \in K_n$, write

$$R = \prod_{i=1}^n r_i, \quad C = \prod_{j=1}^n c_j, \quad P = \text{per } A,$$

where r_i and c_j are the row and column sums. Thus $\Phi(A) = R + C - P$ and

$$\Phi(J_n/n) = 2 - \gamma_n.$$

Theorem 1.1. *For $n \in \{11, 12, 13\}$ and every $A \in K_n$,*

$$\Phi(A) \leq 2 - \gamma_n,$$

with equality if and only if $A = J_n/n$.

We use the following published results.

1. Hwang proved that the only possible positive Φ -maximizer in K_n is J_n/n [5].
2. Cheon and Wanless proved that a partly decomposable matrix cannot maximize Φ , and that a maximizer cannot have its complete zero set equal, up to row and column permutations, to one proper rectangular block [1, Theorems 3.2 and 3.7].
3. Li's cut characterization says that a nonnegative $n \times n$ matrix S is doubly superstochastic if and only if

$$\sum_{i \in I, j \in J} s_{ij} \geq |I| + |J| - n$$

for all $I, J \subseteq [n]$ [6]; see also [1, Lemma 2.2].

4. On a staircase face of the Birkhoff polytope, the barycenter minimizes the permanent [4].
5. The standard averaging method for the face with two prescribed independent zeroes permits a permanent minimizer to be chosen with equal last m rows and equal last m columns; this is recalled and used explicitly by Pula, Song, and Wanless [8, Section 2].
6. The van der Waerden permanent theorem gives $\text{per } B \geq \gamma_n$ for every doubly stochastic B , with equality only at J_n/n [2, 3].

2 Boundary structure and the joint deficit

Because K_n is compact, Φ has a global maximizer. By Hwang's positive-support theorem, a nonuniform maximizer must be a boundary matrix.

Lemma 2.1 (Two independent zeroes). *Every boundary Φ -maximizer has two zero entries in distinct rows and distinct columns.*

Proof. Suppose a nonempty zero set contains no such pair. Fix a zero at (i, j) . Every other zero shares row i or column j . If there were both a zero (i, j') with $j' \neq j$ and a zero (i', j) with $i' \neq i$, those two zeroes would be in distinct rows and columns. Hence all zeroes lie in one row or all lie in one column.

If that row or column is entirely zero, the matrix contains a $1 \times (n-1)$ or $(n-1) \times 1$ zero submatrix and is partly decomposable. Otherwise its complete zero set is one proper rectangular block. Both possibilities are excluded for a maximizer by Cheon–Wanless. Therefore two independent zeroes must occur. $\square$

Assume for contradiction that a boundary maximizer A satisfies

$$\Phi(A) \geq 2 - \gamma_n.$$

Since the row sums and column sums each have arithmetic mean 1, AM–GM gives $R, C \leq 1$. Thus $P \leq \gamma_n$. Write

$$P = \gamma_n - \delta, \quad 0 \leq \delta \leq \gamma_n,$$

and put $R = 1 - \rho$, $C = 1 - \sigma$. The assumed lower bound on Φ gives

$$\rho + \sigma \leq \delta. \tag{1}$$

In particular,

$$R \geq 1 - \delta, \quad C \geq 1 - \delta, \quad RC = (1 - \rho)(1 - \sigma) \geq 1 - \delta. \tag{2}$$

This single shared deficit budget is essential below.

3 The maximal Li scaling and its tight cut

For $I, J \subseteq [n]$, let $s(A[I, J])$ denote the sum of the entries in the indicated submatrix. Define

$$\lambda = \min_{\substack{I, J \subseteq [n] \\ k := |I| + |J| - n > 0}} \frac{s(A[I, J])}{k}. \quad (3)$$

A maximizer is fully indecomposable by the Cheon–Wanless partly-decomposable exclusion. Hence every submatrix occurring in (3) has positive sum, so $\lambda > 0$. Taking $I = J = [n]$ shows $\lambda \leq 1$.

By Li's theorem, A/λ is doubly superstochastic. Therefore there is a doubly stochastic matrix B with

$$\lambda B \leq A \quad \text{entrywise}, \quad P = \text{per } A \geq \lambda^n \text{per } B. \quad (4)$$

Choose I, J attaining the minimum in (3). Put

$$u = n - |I|, \quad v = n - |J|, \quad k = n - u - v > 0.$$

We distinguish the whole cut $(u, v) = (0, 0)$, marginal cuts (exactly one of u, v is zero), and proper cuts $(u, v \geq 1)$.

If $u = v = 0$, then $\lambda = 1$. Since $B \leq A$ and both matrices have total sum n , $A = B$ is doubly stochastic. The van der Waerden theorem gives $P \geq \gamma_n$, and equality would force the positive matrix J_n/n , contradicting that A is a boundary matrix. Hence the whole cut cannot be minimizing.

4 Two elementary deficit estimates

Lemma 4.1 (Subset-sum estimate). *Let $x_1, \dots, x_n > 0$ have sum n and product X . For a subset $S \subseteq [n]$, write*

$$\sum_{i \in S} x_i = |S| + \varepsilon.$$

Then

$$\varepsilon^2 \leq -\frac{n}{2} \log X. \quad (5)$$

Proof. Let $p_i = x_i/n$ and let $q_i = 1/n$. Then

$$D(q\|p) = \sum_i \frac{1}{n} \log \frac{1/n}{x_i/n} = -\frac{1}{n} \log X.$$

Under the two-cell partition S, S^c , the probability difference is $|\varepsilon|/n$. The data-processing inequality and the binary Pinsker inequality give

$$-\frac{1}{n} \log X = D(q\|p) \geq 2 \frac{\varepsilon^2}{n^2},$$

which is (5). □

Lemma 4.2 (A smallest line sum). *If positive $x_1, \dots, x_n$ have sum n , product X , and $\min_i x_i = 1 - a$ with $0 \leq a < 1$, then*

$$a^2 \leq \frac{2(n-1)}{n} (-\log X). \quad (6)$$

Consequently, if $X \geq 1 - \delta$, then

$$a^2 \leq \frac{2(n-1)\delta}{n(1-\delta)}. \quad (7)$$

Proof. AM–GM on the remaining $n - 1$ variables gives

$$X \leq (1-a) \left(1 + \frac{a}{n-1}\right)^{n-1}.$$

Set

$$f(a) = -\log(1-a) - (n-1) \log \left(1 + \frac{a}{n-1}\right).$$

If $g(a) = f(a) - na^2/[2(n-1)]$, then $g(0) = 0$ and

$$g'(a) = \frac{na^2(n-2+a)}{(n-1)(1-a)(n-1+a)} \geq 0.$$

Thus $-\log X \geq f(a) \geq na^2/[2(n-1)]$. The last assertion follows from $-\log(1-\delta) \leq \delta/(1-\delta)$. $\square$

5 The two-independent-zero permanent gap

Let $\Omega_n^{(2)}$ denote the face of the doubly stochastic polytope specified by $b_{12} = b_{21} = 0$.

Proposition 5.1 (Reduction to one variable). *Let $m = n - 2 \geq 2$. The minimum of the permanent on $\Omega_n^{(2)}$ equals the minimum, on*

$$\frac{m-2}{m^2} \leq x \leq \frac{1}{m},$$

of

$$P_n(x) = m!x^{m-2} \left(x^2 a^2 + 2m x a b^2 + m(m-1)b^4 \right), \quad (8)$$

where

$$a = \frac{2-m+m^2x}{2}, \quad b = \frac{1-mx}{2}. \quad (9)$$

Proof. By the averaging method cited above, a permanent minimizer can be chosen in the form

$$Y = \begin{pmatrix} x_1 & 0 & a_1 \mathbf{1}^\top \\ 0 & x_2 & a_2 \mathbf{1}^\top \\ a_1 \mathbf{1} & a_2 \mathbf{1} & x J_m \end{pmatrix}, \quad x_i = 1 - m a_i, \quad a_1 + a_2 + m x = 1.$$

Put $s = a_1 + a_2$ and $p = a_1 a_2$. Apart from the positive factor $m!x^{m-2}$, direct expansion gives

$$E_m(s, p) = x^2(1 - ms + m^2 p) + mx(s^2 - (2 + ms)p) + m(m-1)p^2, \quad x = \frac{1-s}{m}. \quad (10)$$

For fixed s ,

$$\frac{\partial E_m}{\partial p} = 2m(m-1)p + (m+1)s^2 - ms - 1. \quad (11)$$

Since $p \leq s^2/4$ and $0 \leq s \leq 2/m$, the right-hand side is at most

$$D_m(s) = \frac{(m^2 + m + 2)s^2 - 2ms - 2}{2}.$$

The polynomial D_m is convex, while

$$D_m(0) = -1, \quad D_m(2/m) = -1 + \frac{2}{m} + \frac{4}{m^2} < 0 \quad (m \geq 4).$$

Therefore $\partial E_m / \partial p < 0$ throughout the admissible region. The minimum for fixed s occurs at the largest possible p , namely $p = s^2/4$, so $a_1 = a_2 = b$. Solving the doubly stochastic constraints gives (9) and the stated interval for x . Counting the permanent terms according to whether zero, one, or two of the two distinguished diagonal entries are selected gives (8). $\square$

For the three dimensions, choose

$$L_{11} = \frac{113}{800000}, \quad L_{12} = \frac{1083}{20000000}, \quad L_{13} = \frac{10349}{500000000}. \quad (12)$$

Proposition 5.2 (Exact two-zero bounds). *For $n = 11, 12, 13$, every $B \in \Omega_n^{(2)}$ satisfies*

$$\text{per } B > L_n > \gamma_n.$$

Proof. For each n , the exact verifier constructs the polynomial $P_n(x) - L_n \in \mathbb{Q}[x]$. It checks that the polynomial is positive at both endpoints of its complete admissible interval and, by an exact rational Sturm sequence, that it has no root in the interval. Hence it is positive everywhere. The second independent verifier reconstructs P_n by exact Ryser permanent evaluation at $n + 1$ rational points, then repeats the root count with SymPy's independent exact Sturm implementation. The exact outputs are summarized in Table 1. $\square$

6 Marginal tight cuts

Suppose exactly one of u, v is zero. Then the ratio in (3) is the average of a nonempty set of row sums or column sums. Since singleton marginal cuts are also allowed, a minimizing marginal cut has

$$\lambda = \min_i r_i \quad \text{or} \quad \lambda = \min_j c_j.$$

Write $\lambda = 1 - a$. By Lemma 4.2, (2), and $\delta \leq \gamma_n$,

$$a \leq U := \sqrt{\frac{2(n-1)\delta}{n(1-\delta)}} \leq \sqrt{\frac{2(n-1)\delta}{n(1-\gamma_n)}}. \quad (13)$$

For $n = 11, 12, 13$ the exact inequality

$$\frac{2(n-1)}{n} \gamma_n < 1 - \gamma_n$$

holds, so $U < 1$.

The dominated matrix B inherits the two independent zeroes of A . After row and column permutations, Proposition 5.2 applies. Therefore, using Bernoulli's inequality,

$$P \geq \lambda^n \text{per } B > L_n(1 - U)^n, \quad (14)$$

$$L_n(1 - U)^n - (\gamma_n - \delta) \geq L_n - \gamma_n + \delta - nL_n \sqrt{\frac{2(n-1)\delta}{n(1-\gamma_n)}}. \quad (15)$$

Completing the square in $\sqrt{\delta}$ gives the uniform lower bound

$$\eta_n^{\text{marg}} := L_n - \gamma_n - \frac{n(n-1)L_n^2}{2(1-\gamma_n)}. \quad (16)$$

The exact verifier checks $\eta_n^{\text{marg}} > 0$ for all three dimensions. Hence the right-hand side is positive, contradicting $P = \gamma_n - \delta$.

7 Proper tight cuts

Now suppose $u, v \geq 1$. Let

$$\varepsilon_r = \sum_{i \in I^c} r_i - u, \quad \varepsilon_c = \sum_{j \in J^c} c_j - v.$$

Lemma 4.1, followed by Cauchy–Schwarz and (2), yields

$$|\varepsilon_r| + |\varepsilon_c| \leq \sqrt{2(\varepsilon_r^2 + \varepsilon_c^2)} \quad (17)$$

$$\leq \sqrt{-n \log(RC)} \leq \sqrt{\frac{n\delta}{1-\delta}} =: T. \quad (18)$$

The identity

$$s(A[I, J]) = k - \varepsilon_r - \varepsilon_c + s(A[I^c, J^c]) \quad (19)$$

therefore gives

$$\lambda = \frac{s(A[I, J])}{k} \geq 1 - \frac{T}{k}. \quad (20)$$

For the dimensions at hand, $n\gamma_n < 1 - \gamma_n$, so $T < 1 \leq k$.

Because B is doubly stochastic,

$$s(B[I, J]) = k + s(B[I^c, J^c]) \geq k.$$

On the other hand, tightness and (4) give

$$\lambda s(B[I, J]) \leq s(A[I, J]) = \lambda k.$$

Thus equality holds and

$$B[I^c, J^c] = 0. \quad (21)$$

So B lies on the face with a prescribed $u \times v$ zero block.

After permuting rows and columns, this is a two-step staircase face. Hwang's theorem says that its barycenter minimizes the permanent. The barycenter is

$$H_{n;u,v} = \begin{pmatrix} \frac{1}{n-v} J_{u,n-v} & 0_{u,v} \\ \frac{k}{(n-u)(n-v)} J_{n-u,n-v} & \frac{1}{n-u} J_{n-u,v} \end{pmatrix}. \quad (22)$$

There are $(n-v)!(n-u)!/k!$ supported permutations, each of the same weight. Hence

$$\text{per } B \geq \nu_{n;u,v} := \text{per } H_{n;u,v} = \frac{\gamma_{n-u} \gamma_{n-v}}{\gamma_k}, \quad k = n - u - v, \quad (23)$$

where $\gamma_0 := 1$.

Using (20), Bernoulli's inequality, and $1 - \delta \geq 1 - \gamma_n$,

$$\nu_{n;u,v} \left(1 - \frac{T}{k}\right)^n - (\gamma_n - \delta) \geq \nu_{n;u,v} - \gamma_n + \delta - \frac{n\nu_{n;u,v}}{k} \sqrt{\frac{n\delta}{1-\gamma_n}}. \quad (24)$$

Completing the square gives

$$\eta_{n;u,v} := \nu_{n;u,v} - \gamma_n - \frac{n^3 \nu_{n;u,v}^2}{4k^2(1-\gamma_n)}. \quad (25)$$

The verifier checks $\eta_{n;u,v} > 0$ for every pair

$$u, v \geq 1, \quad u + v \leq n - 1,$$

namely 45, 55, and 66 proper cuts for $n = 11, 12, 13$. This again contradicts $P = \gamma_n - \delta$.

The whole, marginal, and proper cases exhaust every tight cut. Thus a boundary maximizer cannot exist. Hwang's positive-support theorem identifies the sole remaining maximizer as J_n/n , proving Theorem 1.1.

8 Exact computational certificate

No floating-point value is used in a correctness decision. The primary verifier uses Python's standard-library `Fraction` class and checks:

1. the two-variable equalization identity (11);
2. the exact polynomial (8);
3. positivity of $P_n - L_n$ at both interval endpoints;
4. zero roots in the interval by an exact rational Sturm sequence;
5. $L_n > \gamma_n$ and $\eta_n^{\text{marg}} > 0$;
6. the barycenter permanent formula (23);
7. $\eta_{n;u,v} > 0$ for every proper cut.

The independent audit does not import the primary verifier. It uses exact SymPy polynomials and Sturm sequences, and it validates P_n at $n + 1$ distinct rational points by a literal exact Ryser permanent calculation. Since every entry of the matrix is affine in x , its permanent has degree at most n ; agreement at $n + 1$ points proves the polynomial identity.

Table 1: Exact certificate summary. Decimal values are displayed only for readability.

n	L_n	$L_n - \gamma_n$	η_n^{marg}	$\min \eta_{n;u,v}$	minimizing (u, v, k)
11	113/800000	$1.3440511 \cdot 10^{-6}$	$2.4656165 \cdot 10^{-7}$	$5.9999275 \cdot 10^{-7}$	(1, 1, 9)
12	1083/20000000	$4.2678291 \cdot 10^{-7}$	$2.3324583 \cdot 10^{-7}$	$2.0401292 \cdot 10^{-7}$	(1, 1, 10)
13	10349/500000000	$1.3830175 \cdot 10^{-7}$	$1.0488530 \cdot 10^{-7}$	$6.7829049 \cdot 10^{-8}$	(1, 1, 11)

For the Sturm checks, the complete admissible intervals are

$$\left[\frac{7}{81}, \frac{1}{9} \right], \quad \left[\frac{2}{25}, \frac{1}{10} \right], \quad \left[\frac{9}{121}, \frac{1}{11} \right]$$

for $n = 11, 12, 13$, respectively. In each case both endpoint values of $P_n - L_n$ are positive and the exact root count is zero.

9 Scope and remaining range

The proof above establishes the three dimensions 11, 12, 13. It does not by itself address $5 \leq n \leq 10$. The scalar marginal inequality (16) is the bottleneck when the same constants and face bound are used below $n = 11$; a sharper marginal estimate or a different permanent gap is required there.

As of the cited June 2026 preprint, Pang proves the conjecture for $n \geq 17$ [7]. The present note should be treated as a new, exact computer-assisted working proof until it has been independently reviewed.

Reproducibility

Run

```
python3 -I verify_dittert_n11_n13.py
python3 -I audit_dittert_n11_n13_sympy.py
```

The expected final lines are

```
ALL CASES CERTIFIED
INDEPENDENT AUDIT CERTIFIED
```

References

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